Chapter 13: ANOVA

ENGR 0021

Review

A regression line for predicting test scores has an MSE of 64 points.

(a) About 68% of the time, the predictions will be right to within _____ points.

(b) About 95% of the time, the predictions will be right to within _____ points.

Why ANOVA?

- We want to compare means across multiple groups
- Example: surface vs middepth vs bottom
- With 2 groups → use a t-test
- With 3+ groups → need a new approach

- Key idea:

  Compare variation between groups vs within groups

ANOVA Example

- The Wolf River in Tennessee flows past an abandoned site once used by the pesticide industry for dumping wastes, including chlordane (pesticide), aldrin, and dieldrin (both insecticides).
- These highly toxic organic compounds can cause various cancers and birth defects.
- The standard methods to test whether these substances are present in a river is to take samples at six-tenths depth.
- But since these compounds are denser than water and their molecules tend to stick to particles of sediment, they are more likely to be found in higher concentrations near the bottom than near mid-depth.

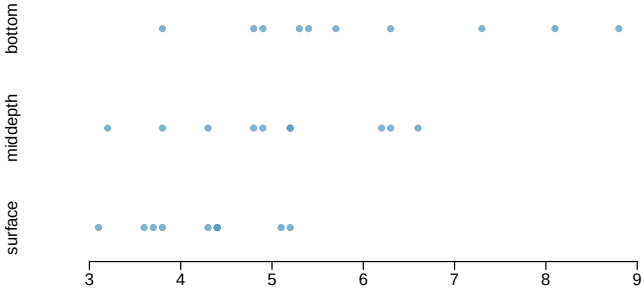
Data

Aldrin concentration (nanograms per liter) at three levels of depth.

	aldrin	depth
1	3.80	bottom
2	4.80	bottom
...		
10	8.80	bottom
11	3.20	middepth
12	3.80	middepth
...		
20	6.60	middepth
21	3.10	surface
22	3.60	surface
...		
30	5.20	surface

Exploratory analysis

Aldrin concentration (nanograms per liter) at three levels of depth.



	n	mean	sd
bottom	10	6.04	1.58
middepth	10	5.05	1.10
surface	10	4.20	0.66
overall	30	5.1 0	1.37

Key Idea of ANOVA

- Do the group means differ?
- Look at two sources of variation:
  - Between groups (signal)
  - Within groups (noise)

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

Connection to Regression

- ANOVA and regression use the same idea:

$$\text{Total variation} = \text{explained} + \text{unexplained}$$

- Regression: $SST = SSR + SSE$
- ANOVA: $SST = SSG + SSE$

Research question

Is there a difference between the mean aldrin concentrations among the three levels?

- To compare means of 2 groups we use a Z or a T statistic.
- To compare means of 3+ groups we use a new test called **ANOVA** and **F** statistic.

Conditions

1. The observations should be independent within and between groups
   - If the data are a simple random sample from less than 10% of the population, this condition is satisfied.
   - Carefully consider whether the data may be independent (e.g. no pairing).
   - Always important, but sometimes difficult to check.
2. The observations within each group should be nearly normal.
   - Especially important when the sample sizes are small.

How do we check for normality?

3. The variability across the groups should be about equal.
   - Especially important when the sample sizes differ between groups.

How can we check this condition?

z/t test vs. ANOVA - Purpose

z/t test

Compare means from **two** groups to see whether they are so far apart that the observed difference cannot reasonably be attributed to sampling variability.

$$H_0 : \mu_1 = \mu_2$$

ANOVA

Compare the means from **two or more** groups to see whether they are so far apart that the observed differences cannot all reasonably be attributed to sampling variability.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

z/t test vs. ANOVA - Method

z/t test

Compute a test statistic (a ratio).

$$z/t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{SE(\bar{x}_1 - \bar{x}_2)}$$

ANOVA

Compute a test statistic (a ratio).

$$F = \frac{\text{variability bet. groups}}{\text{variability w/in groups}}$$

- Large test statistics lead to small p-values.
- If the p-value is small enough H_0 is rejected, we conclude that the population means are not equal.

z/t test vs. ANOVA

- With only two groups t-test and ANOVA are equivalent, but only if we use a pooled standard variance in the denominator of the test statistic.
- With more than two groups, ANOVA compares the sample means to an overall **grand mean**.

Hypotheses

What are the correct hypotheses for testing for a difference between the mean aldrin concentrations among the three levels?

(a) $H_0 : \mu_B = \mu_M = \mu_S$
$H_A : \mu_B \neq \mu_M \neq \mu_S$

(b) $H_0 : \mu_B \neq \mu_M \neq \mu_S$
$H_A : \mu_B = \mu_M = \mu_S$

(c) $H_0 : \mu_B = \mu_M = \mu_S$
H_A : At least one mean is different.

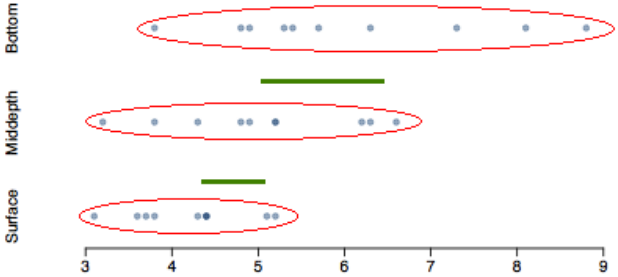
(d) $H_0 : \mu_B = \mu_M = \mu_S = 0$
H_A : At least one mean is different.

(e) $H_0 : \mu_B = \mu_M = \mu_S$
$H_A : \mu_B > \mu_M > \mu_S$

Test statistic

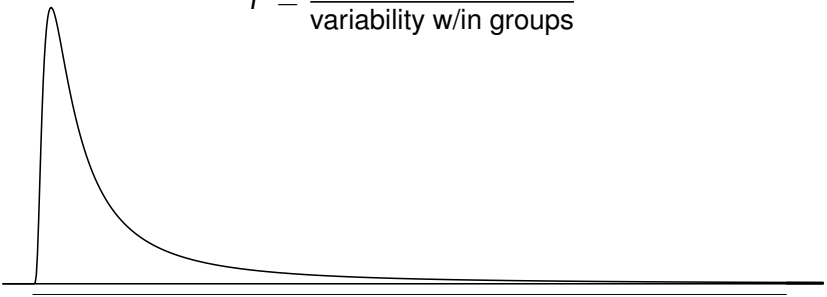
Does there appear to be a lot of variability within groups? How about between groups?

$$F = \frac{\text{variability bet. groups}}{\text{variability w/in groups}}$$



F distribution and p-value

$$F = \frac{\text{variability bet. groups}}{\text{variability w/in groups}}$$



- In order to be able to reject H_0, we need a small p-value, which requires a large F statistic.
- In order to obtain a large F statistic, variability between sample means needs to be greater than variability within sample means.

Df

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	8.48	6.13	0.0063
(**E**rror)	Residuals	27	37.33	1.38		
	Total	29	54.29			

Degrees of freedom associated with ANOVA

- groups:
- total:
- error:

SSG

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	**16.96**	8.48	6.13	0.0063
(**E**rror)	Residuals	27	37.33	1.38		
	Total	29	54.29			

Sum of squares between groups, SSG

Measures the variability between groups

$$SSG = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2$$

where n_i is each group size, $\bar{x}_i$ is the average for each group, $\bar{x}$ is the overall (grand) mean.

	n	mean
bottom	10	6.04
middepth	10	5.05
surface	10	4.2
overall	30	5.1

Sum of squares total, SST

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	8.48	6.13	0.0063
(**E**rror)	Residuals	27	37.33	1.38		
	Total	29	**54.29**			

Measures the total variability between groups

$$SST = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

where x_i represent each observation in the dataset.

Sum of squares error, SSE

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	8.48	6.13	0.0063
(**E**rror)	Residuals	27	**37.33**	1.38		
	Total	29	54.29			

Measures the variability within groups:

$$SSE = SST - SSG$$

Mean Squared Error (MSE)

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	**8.48**	6.13	0.0063
(**E**rror)	Residuals	27	37.33	**1.38**		
	Total	29	54.29			

Mean square error is calculated as sum of squares divided by the degrees of freedom.

F-Statistic

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	8.48	**6.14**	0.0063
(**E**rror)	Residuals	27	37.33	1.38		
	Total	29	54.29			

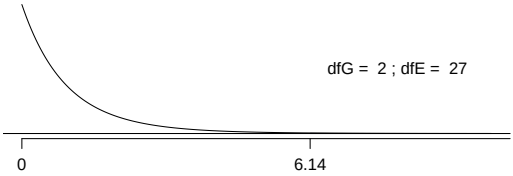
As we discussed before, the F statistic is the ratio of the between group and within group variability.

$$F = \frac{MSG}{MSE}$$

p-value

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	8.48	**6.14**	0.0063
(**E**rror)	Residuals	27	37.33	1.38		
	Total	29	54.29			

p-value is the probability of at least as large a ratio between the "between group" and "within group" variability, if in fact the means of all groups are equal. It's calculated as the area under the F curve, with degrees of freedom df_G and df_E, above the observed F statistic.



Conclusion - in context

What is the conclusion of the hypothesis test?

The data provide convincing evidence that the average aldrin concentration

(a) is different for all groups.

(b) on the surface is lower than the other levels.

(c) is different for at least one group.

(d) is the same for all groups.

Conclusion

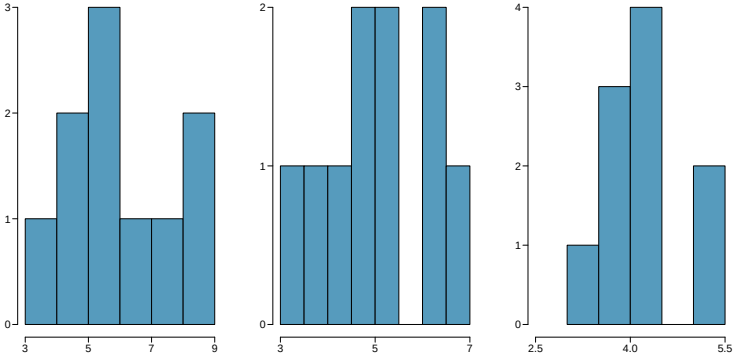
- If p-value is small (less than α), reject H_0. The data provide convincing evidence that at least one mean is different from (but we can't tell which one).
- If p-value is large, fail to reject H_0. The data do not provide convincing evidence that at least one pair of means are different from each other, the observed differences in sample means are attributable to sampling variability (or chance).

(1) independence

Does this condition appear to be satisfied?

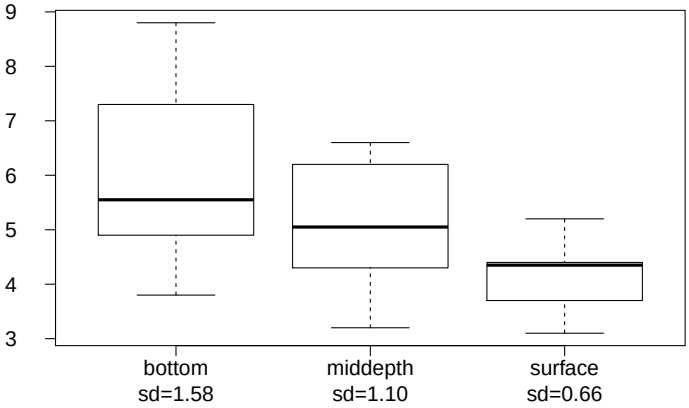
(2) approximately normal

Does this condition appear to be satisfied?



(3) constant variance

Does this condition appear to be satisfied?



Which means differ?

- Earlier we concluded that at least one pair of means differ. The natural question that follows is “which ones?”
- We can do two sample t tests for differences in each possible pair of groups.

Can you see any pitfalls with this approach?

- When we run too many tests, the Type 1 Error rate increases.
- This issue is resolved by using a modified significance level.

Multiple comparisons

- The scenario of testing many pairs of groups is called **multiple comparisons**.
- The **Bonferroni correction** suggests that a more **stringent** significance level is more appropriate for these tests:

$$\alpha^\star = \alpha / K$$

  where K is the number of comparisons being considered.
- If there are k groups, then usually all possible pairs are compared and $K = \frac{k(k-1)}{2}$.

Determining the modified α

In the aldrin data set depth has 3 levels: bottom, mid-depth, and surface. If $\alpha = 0.05$, what should be the modified significance level for two sample t tests for determining which pairs of groups have significantly different means?

(a) $\alpha^* = 0.05$

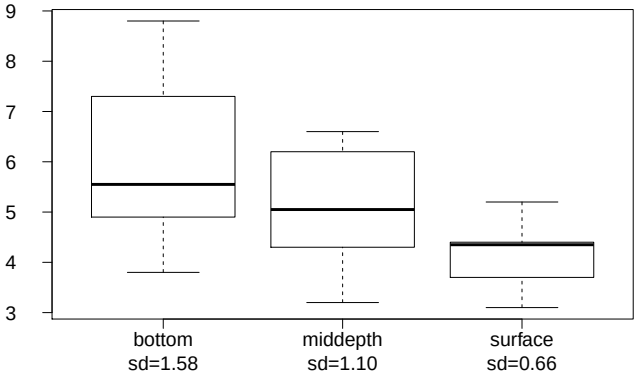
(b) $\alpha^* = 0.05/2 = 0.025$

(c) $\alpha^* = 0.05/3 = 0.0167$

(d) $\alpha^* = 0.05/6 = 0.0083$

Which means differ?

Based on the box plots below, which means would you expect to be significantly different?



(a) bottom & surface

(b) bottom & mid-depth

(c) mid-depth & surface

(d) bottom & mid-depth; mid-depth & surface

(e) bottom & mid-depth; bottom & surface; mid-depth & surface

Which means differ? (cont.)

If the ANOVA assumption of equal variability across groups is satisfied, we can use the data from all groups to estimate variability:

- Estimate any within-group standard deviation with $\sqrt{MSE}$, which is s_{pooled}
- Use the error degrees of freedom, $n - k$, for t-distributions

Difference in two means: after ANOVA

$$SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \approx \sqrt{\frac{MSE}{n_1} + \frac{MSE}{n_2}}$$

Pairwise Comparisons

Is there a difference between the average aldrin concentration at the bottom and at mid depth?

	n	mean	sd
bottom	10	**6.04**	1.58
middepth	10	**5.05**	1.10
surface	10	4.2	0.66
overall	30	5.1	1.37

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
depth	2	16.96	8.48	6.13	0.0063
Residuals	**27**	37.33	**1.38**		
Total	29	54.29			

$$T_{df_E} = \frac{(\bar{x}_{bottom} - \bar{x}_{middepth})}{\sqrt{\frac{MSE}{n_{bottom}} + \frac{MSE}{n_{middepth}}}}$$

Fail to reject H_0, data do not provide convincing evidence of a difference between average aldrin concentrations at bottom and mid depth.

Pairwise Comparisons

Is there a difference between the average aldrin concentration at the bottom and at surface?

$$T_{df_E} = \frac{(\bar{x}_{bottom} - \bar{x}_{surface})}{\sqrt{\frac{MSE}{n_{bottom}} + \frac{MSE}{n_{surface}}}}$$

ANOVA Summary

- Goal: Compare means across multiple groups

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{vs} \quad H_A : \text{At least one mean differs}$$

- Key idea: **signal vs noise**

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

- Interpretation:
  - Large $F \rightarrow$ strong evidence against H_0
  - Small p-value $\rightarrow$ reject H_0
- Important notes:
  - ANOVA tells us **at least one mean differs**
  - It does **not** tell us which groups differ
  - Follow-up: pairwise comparisons (with correction)
- With two groups:

$$\text{ANOVA} \equiv \text{t-test}$$